Name: ______________________

TRANSFORMATIONS, (TRANSLATION)

Date: _________________

LO: I can translate shapes using column vectors and describe translations.

Translation

A **translation** slides every point of a shape the same distance in the same direction. It is described by a **column vector**.

Column vector: The top number is horizontal movement (+ right, - left). The bottom is vertical (+ up, - down).

Quick examples

	Vector (3, -2): right 3, down 2	positive x, negative y
	(2,5) translated by (3,1) = (5,6)	add vector
	Translation preserves size, shape and orientation	no rotation
	Vector (3, -2) means left 3, up 2	positive = right/ up
	Translation changes the size of the	only position

Key idea

Add the column vector to each point: new point = original + vector.

(x, y) + (a, b) = (x+a, y+b)

a = horizontal shift, b = vertical shift

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – applying a vector

Translate (3, 2) by the vector (4, -1).

Add horizontal: 3 + 4 = 7

Add vertical: 2 + (-1) = 1

Image: (7, 1)

(7, 1)

Add each component of the vector to the corresponding coordinate.

Example 2 – negative vector

Translate (5, 3) by the vector (-2, -4).

Add horizontal: 5 + (-2) = 3

Add vertical: 3 + (-4) = -1

Image: (3, -1)

(3, -1)

Negative components move left or down.

Example 3 – describing a translation

(1, 4) is mapped to (5, 2). Describe the translation.

Horizontal: 5 - 1 = 4 (right)

Vertical: 2 - 4 = -2 (down)

Vector: (4, -2)

Translation by vector (4, -2)

Subtract original from image to find the vector.

Example 4 – translating a shape

Triangle (1,1), (3,1), (2,4) translated by (-3, 2).

(1,1) (-2, 3)

(3,1) (0, 3)

(2,4) (-1, 6)

(-2,3), (0,3), (-1,6)

Apply the same vector to every vertex.

YOUR TURN – PRACTICE (deliberate practice)

1. Applying vectors

Translate each point by the given vector.

a) (2, 3) by (4, 1)

b) (5, 1) by (-3, 2)

c) (-1, 4) by (3, -5)

d) (0, 0) by (-2, -3)

2. Finding the vector

Find the translation vector from the first point to the second.

a) (1, 2) to (4, 5)

b) (5, 3) to (2, 7)

c) (3, -1) to (-2, 4)

d) (-4, 6) to (1, 1)

3. Translating shapes

Translate each set of vertices by vector (2, -3).

a) (1,4), (3,4), (2,6)

b) (0,5), (4,5), (4,7), (0,7)

c) (2,1), (5,1), (5,3)

d) (-1,3), (2,3), (2,5), (-1,5)

4. Reverse translations

Find the original point before the translation.

a) Image (7, 2), vector (3, -1). Original?

b) Image (1, 5), vector (-4, 2). Original?

c) Image (-2, 0), vector (-5, -3). Original?

d) Image (6, -1), vector (2, -4). Original?

5. Mixed translation problems

Solve each problem.

a) (3, 7) translated by (-5, 3)

b) Vector to map (2, -1) to (6, 4)?

c) After vector (a, 3), (1, 2) maps to (4, 5). Find a.

d) Two translations: (2, 1) then (3, -4). Combined vector?

e) Reverse of vector (3, -2)?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) A translation is described by a column vector ☐

Correct answer: __________

Reason: __________________

b) A positive x-component means move left ☐

Correct answer: __________

Reason: __________________

c) Translation preserves size, shape and orientation ☐

Correct answer: __________

Reason: __________________

d) Translating by (0, 0) moves the shape to the origin ☐

Correct answer: __________

Reason: __________________

e) To find the vector, subtract original from image ☐

Correct answer: __________

Reason: __________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A shape is translated by vector **(3, -2)** and then by vector **(-5, 4)**. Find the single vector that has the same effect as both translations combined.

Expression: ______________________________

Simplified: ______________________________

b) Point A(2, 1) is translated to A'(5, 4). Point B is translated by the same vector to B'(1, 7). Find the coordinates of B and the length of AB.

Expression: ______________________________

Simplified: ______________________________

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!